

Cambridge Lower Secondary Checkpoint

MATHEMATICS**0862/02**

Paper 2

April 2025

MARK SCHEME

Maximum Mark: 50

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Markers were instructed to award marks. It does not indicate the details of the discussions that took place at a Markers' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the End of Series Report. Cambridge will not enter into discussions about these mark schemes.

This document has **10** pages. Any blank pages are indicated.

Mark scheme annotations and abbreviations

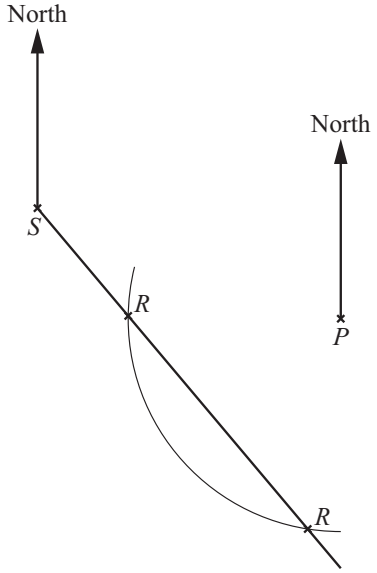
FT	follow through after error
SC	special case mark
cao	correct answer only
dep	dependent
isw	ignore subsequent working
nfw	not from wrong working
oe	or equivalent
soi	seen or implied

Question	Answer	Marks	Part marks	Guidance												
1	32 (mins)	1														
2	14	1		Ignore decimals in working if 14 chosen as answer. Accept ± 14												
3	$(n + 7)^2$	1		Accept equivalents, e.g. $(n + 7)(n + 7)$, $n^2 + 14n + 49$ Do not isw incorrect expansion. Do not accept an equation, e.g. $x = (n + 7)^2$, $n = (n + 7)^2$												
4	2	1		Accept any clear indication.												
5	343 (cm ³) or 343.0 to 343.4... (cm ³)	2	Award 1 mark for $\pi \times 4.1^2$ [$\times 6.5$] oe.	Accept values of π between 3.14 and $\frac{22}{7}$ 1 mark implied by 52.7 to 52.8...												
6	66 and 3	2	Award 1 mark for each correct answer.													
7	<table><tr><td></td><td>Meat</td><td>Vegetarian</td></tr><tr><td>Children under 10</td><td>(2)</td><td>(3)</td></tr><tr><td>Children 10 to 17</td><td>7</td><td>3</td></tr><tr><td>Adults</td><td>3</td><td>5</td></tr></table>		Meat	Vegetarian	Children under 10	(2)	(3)	Children 10 to 17	7	3	Adults	3	5	2	Award 1 mark for two correct values.	
	Meat	Vegetarian														
Children under 10	(2)	(3)														
Children 10 to 17	7	3														
Adults	3	5														
8	7.2 (yuan per dollar)	1														
9	(0, 4)	1														

Question	Answer	Marks	Part marks	Guidance																				
10(a)	A correct, ordered diagram <table border="1" data-bbox="331 347 656 507"> <tr> <td colspan="4">(Question 1)</td> <td></td> </tr> <tr> <td></td><td></td><td></td><td>4</td> <td>(3)</td> </tr> <tr> <td>7</td><td>6</td><td>1</td><td>1</td> <td>(4)</td> </tr> <tr> <td></td><td></td><td>1</td><td>0</td> <td>(5)</td> </tr> </table>	(Question 1)								4	(3)	7	6	1	1	(4)			1	0	(5)	2	Award 1 mark for all rows correct but unordered or for correctly ordered but with one error or omission.	For 1 or 2 marks ignore numbers added to the right-hand side of the stem-and-leaf diagram. One error could be, e.g. one leaf in the wrong row.
(Question 1)																								
			4	(3)																				
7	6	1	1	(4)																				
		1	0	(5)																				
10(b)	(Median) 43 (Mode) 36	2	Award 1 mark for each correct answer.																					
10(c)	range and lower	1		Both words in the correct order for the mark.																				
11	144(°)	2	Award 1 mark for $(10 - 2) \times 180 [\div 10]$ or better or $180 - \frac{360}{10}$ oe.	Or better, e.g. 1440																				
12(a)	375 ($\leq n < 385$)	1																						
12(b)	3.625 (kg)	1		Do not accept with an inequality sign, e.g. < 3.625																				
13	900	2	Award 1 mark for $750 \div (11 - 6) \times k$ or better where $k = 1, 6, 11$ or 17	Or better, e.g. $750 \div 5$ Implied by 150, 1650 or 2550																				

Question	Answer	Marks	Part marks	Guidance										
14	Could be correct ☒ ☐ Not correct ☐ ☒	1		Both answers correct for the mark. Accept any clear indication.										
15	(x =) 1.04	1		Accept equivalent fractions, e.g. $\frac{104}{100}$, $\frac{26}{25}$ Do not accept 1.04^5 or 104%.										
16	<table><tr><th>True</th><th>False</th></tr><tr><td>☒</td><td>☐</td></tr><tr><td>☒</td><td>☐</td></tr><tr><td>☐</td><td>☒</td></tr><tr><td>☒</td><td>☐</td></tr></table>	True	False	☒	☐	☒	☐	☐	☒	☒	☐	2	Award 1 mark for three correct answers.	Accept any clear indication.
True	False													
☒	☐													
☒	☐													
☐	☒													
☒	☐													
17	0.59 oe	3	Award 1 mark for $P(E) = 0.07 \times 2$ oe and Award 1 mark for $1 - (0.06 + \textit{their } P(E) + 0.07 + 0.1 + 0.04)$ oe or $0.73 - \textit{their } P(E)$ correctly evaluated for <i>their</i> $P(E)$ provided <i>their</i> $P(E) < 0.73$	Accept any equivalent fraction or 59%. If answer and table contradict, mark answer line. Implied by $P(E) = 0.14$ oe or sight of 0.41 oe. oe, e.g. $1 - 0.41$, $1 - (0.27 + \textit{their } P(E))$.										

Question	Answer	Marks	Part marks	Guidance									
18	$x > 6.4$ oe	2	Award 1 mark for correctly isolating terms in x on one side and constants on the other side, e.g. $11x - 6x > 32$ or better.	Accept equivalent fractions, e.g. $x > \frac{32}{5}, x > 6\frac{2}{5}$ for 2 marks. Accept $-32 > 6x - 11x$ or better. Or better, e.g. $5x > 32, -32 > -5x$ For 1 mark, accept any inequality sign or equals sign. Do not isw. Answer $x = 6.4$ or 6.4 scores 1 mark only.									
19	<table><tr><td></td><td>Positive y-coordinate</td><td>Negative y-coordinate</td></tr><tr><td>Positive x-coordinate</td><td>(B')</td><td>C'</td></tr><tr><td>Negative x-coordinate</td><td>A'</td><td>D'</td></tr></table>		Positive y-coordinate	Negative y-coordinate	Positive x-coordinate	(B')	C'	Negative x-coordinate	A'	D'	2	Award 1 mark for two correct answers or for $A'B'C'D'$ drawn correctly on the grid.	Ignore diagram if table correct. Accept only the points $(-3, 1), (3, 1), (1, -1), (-1, -5)$ drawn on the grid. Mark intention.
	Positive y-coordinate	Negative y-coordinate											
Positive x-coordinate	(B')	C'											
Negative x-coordinate	A'	D'											

Question	Answer	Marks	Part marks	Guidance
20	Both positions of R correctly marked, e.g. 	3	Award 2 marks for one correct point for R or If 2 marks not awarded: Award 1 mark for the correct bearing of 135° from S drawn and Award 1 mark for 3.5 (cm) soi.	Ignore line joining S to P. Tolerance $\pm 2^\circ$ and ± 2 mm. Ignore any incorrect points for these 2 marks. Tolerance $\pm 2^\circ$. If the value 3.5 is not written then: 3.5 is implied by a correct arc from P. If arc not seen 3.5 can be implied by all <i>their</i> points for R being 3.5 cm from P. Tolerance ± 2 mm for the drawing but not the written value. If no intersecting bearing and arc to indicate the position of R clearly there must be a point marked, not just the label.
21	$x^2 + 6x + 9$	2	Award 1 mark for three correct terms out of x^2 , $3x$, $3x$, 9	Note: $6x$ counts as two correct terms.

Question	Answer	Marks	Part marks	Guidance
22	1734 (cm ²)	2	Award 1 mark for $16^2 + 30^2$ or better or for <i>their</i> 34×51 where <i>their</i> 34 comes from an attempt at Pythagoras.	Or better, e.g. $256 + 900$, 1156, $\sqrt{16^2 + 30^2}$, 34 e.g. $\sqrt{30^2 - 16^2} \times 51$
23	(x =) 2.1 oe	3	Award 2 marks for a correct equation which does not have x in a denominator and which has no brackets, e.g. $27 = 20x - 15$ or $\frac{27}{5} = 4x - 3$ or better or Award 1 mark for $27 = 5(4x - 3)$	Accept equivalent fractions for 3 marks, e.g. $\frac{42}{20}$, $\frac{21}{10}$, $2\frac{1}{10}$ but not, e.g. $\frac{8.4}{4}$ as the fraction cannot contain decimals for 3 marks. Or better, e.g. $42 = 20x$, $5.4 = 4x - 3$ Award 2 marks for an embedded answer.
24	(t =) 2y	2	Award 1 mark for $t - y = \frac{t+y}{3}$ oe or better or for answer (t =) $-2y$	Do not accept $t = \frac{4y}{2}$ Or better, e.g. $3(t - y) = t + y$, $3t - 3y = t + y$, $2t = 4y$

Question	Answer	Marks	Part marks	Guidance
25	83.4 (cm) cao	4	Award 3 marks for any of these answers • 83 • 83.3 to 83.43 • 75.4 • $24\pi + 8$ or Award 2 marks for $\frac{\pi \times 20}{2} + \frac{\pi \times (20+8)}{2} [+8] \text{ oe or better}$ or Award 1 mark for any of these • $\frac{\pi \times 20}{2}$ oe or better • $\frac{\pi \times (20+8)}{2}$ oe or better or Award 1 mark for rounding <i>their</i> more accurate answer shown correct to 1 decimal place.	Accept values of π between 3.14 and $\frac{22}{7}$ Or better, e.g. $10\pi + 14\pi$, (31.4 to 31.43) + (43.9 to 44) 24π, 75 or 75.3 to 75.43 If 8 added to one of the arcs first then $\frac{\pi \times 20}{2} + 8$ implied by 39.4 to 39.43 $\frac{\pi \times (20+8)}{2} + 8$ implied by 51.9 to 52 Or better, e.g. 10π, 31.4 to 31.43 Or better, e.g. 14π, 43.9 to 44

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